Cambridge (CIE) IGCSE Additional Maths

Your notes

Quadratic Functions

Contents

Your notes

Solving Quadratics by Factorising

How do I solve a quadratic equation using factorisation?

- Rearrange it into the form $ax^2 + bx + c = 0$
  - zero must be on one side
    - it is easier to use the side where a is positive
- Factorise the quadratic and solve each bracket equal to zero
  - If $(x + 4)(x - 1) = 0$, then either $x + 4 = 0$ or $x - 1 = 0$
    - Because if $A \times B = 0$, then either $A = 0$ or $B = 0$
- To solve $(x-3)(x+7)=0$
  - ...solve "**first bracket = 0**":
    - $x - 3 = 0$
    - add 3 to both sides: $x = 3$
  - ...and solve "**second bracket = 0**"
    - $x + 7 = 0$
    - subtract 7 from both sides: $x = -7$
  - The **two solutions** are $x = 3$ or $x = -7$
    - The solutions have the opposite signs to the numbers in the brackets
- To solve $(2x-3)(3x+5)=0$
  - ...solve "first bracket = 0"
    - $2x - 3 = 0$
    - add 3 to both sides: $2x = 3$
    - divide both sides by 2: $x = \frac{3}{2}$
  - ...solve "second bracket = 0"

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- $3x + 5 = 0$
- subtract 5 from both sides: $3x = -5$
- divide both sides by 3: $x = -\frac{5}{3}$

- The two solutions are $x = \frac{3}{2}$ or $x = -\frac{5}{3}$

- To solve $x(x-4) = 0$
  - it may help to think of x as (x – 0) or (x)
  - ...solve "first bracket = 0"
    - (x) = 0, so $x = 0$
  - ...solve "second bracket = 0"
    - $x - 4 = 0$
    - add 4 to both sides: $x = 4$
  - The **two solutions** are $x = 0$ or $x = 4$
    - It is a common **mistake** to divide both sides by x at the beginning - you will **lose a solution** (the $x = 0$ solution)

Examiner Tips and Tricks

- Where permitted, and if you calculator has a quadratic solving feature, you can use it to check your final solutions!
  - Such calculators also help you to factorise (if you're struggling with that step)
  - e.g. A calculator gives solutions to $6x^2 + x - 2 = 0$ as $x = -\frac{2}{3}$ and $x = \frac{1}{2}$
    - "Reverse" the method above to factorise!
    - $6x^2 + x - 2 = (3x + 2)(2x - 1)$
  - Warning: a calculator (correctly) gives solutions to $12x^2 + 2x - 4 = 0$ as $x = -\frac{2}{3}$ and $x = \frac{1}{2}$
    - But $12x^2 + 2x - 4 \neq (3x+2)(2x-1)$ as these brackets expand to $6x^2 + \ldots$ not $12x^2 + \ldots$
    - Multiply by 2 to correct this

- $12x^2 + 2x - 4 = 2(3x+2)(2x-1)$

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Worked Example

(a) Solve $(x-2)(x+5)=0$

Set the first bracket equal to zero

$$x - 2 = 0$$

Add 2 to both sides

$$x = 2$$

Set the second bracket equal to zero

$$x + 5 = 0$$

Subtract 5 from both sides

$$x = -5$$

Write both solutions together using "or"

$$\boldsymbol{x = 2 \text{ or } x = -5}$$

(b) Solve $(8x+7)(2x-3)=0$

Set the first bracket equal to zero

$$8x + 7 = 0$$

Subtract 7 from both sides

$$8x = -7$$

Divide both sides by 8

$$x = -\frac{7}{8}$$

Set the second bracket equal to zero

Your notes

$$2x - 3 = 0$$

Add 3 to both sides

$$2x = 3$$

Divide both sides by 2

$$x = \frac{3}{2}$$

Write both solutions together using "or"

$$x = -\frac{7}{8} \text{ or } x = \frac{3}{2}$$

(c) Solve $x(5x - 1) = 0$

Do not divide both sides by x (this will lose a solution at the end)
Set the first "bracket" equal to zero

$$(x) = 0$$

Solve this equation to find x

$$x = 0$$

Set the second bracket equal to zero

$$5x - 1 = 0$$

Add 1 to both sides

$$5x = 1$$

Divide both sides by 5

$$x = \frac{1}{5}$$

Write both solutions together using "or"

$$x = 0 \text{ or } x = \frac{1}{5}$$

Quadratic Formula

Your notes

Quadratic Formula

How do I use the quadratic formula to solve a quadratic equation?

- A quadratic equation has the form:

  $ax^2 + bx + c = 0$ (as long as $a \neq 0$)

  - you need "= 0" on one side
- The **quadratic formula** is a formula that gives both solutions:
  - $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- Read off the values of a, b and c from the equation
- **Substitute** these into the formula
  - write this line of working in the exam
  - Put **brackets** around any **negative numbers** being substituted in
- To solve $2x^2 - 5x + 2 = 0$ using the quadratic formula:
  - $a = 2$, $b = -5$ and $c = 2$
  - $x = \dfrac{-(-5) \pm \sqrt{(-5)^2 - 4 \times 2 \times 2}}{2 \times 2}$
  - Carefully simplify by doing the calculation in parts $x = \dfrac{5 \pm \sqrt{25 - 16}}{4} = \dfrac{5 \pm \sqrt{9}}{4}$
  - Separate the + and - to get the two answers $x = \dfrac{5+3}{4} = \dfrac{8}{4} = 2$, $x = \dfrac{5-3}{4} = \dfrac{2}{4} = \dfrac{1}{2}$
- On the **non-calculator** paper, answers may be required in **exact** form

- On the **calculator** paper, you might have to round your answers
  - Give your answers in exact form before rounding in case you round incorrectly

Your notes

Examiner Tips and Tricks

- On the **calculator** paper you will be able to check your solutions using the quadratic equation solver feature if your calculator has one
- Always look for how the question wants you to leave your final answers
  - for example, correct to 2 decimal places

Worked Example

Use the quadratic formula, without a calculator, to find the exact solutions of the equation $12x^2 - 17x + 6 = 0$.

Write down the values of *a*, *b* and c

$$a = 12,\ b = -17,\ c = 6$$

Substitute these values into the quadratic formula, $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Put brackets around any negative numbers

$$x = \frac{-(-17) \pm \sqrt{(-17)^2 - 4 \times 12 \times 6}}{2 \times 12}$$

Simplify

$$x = \frac{17 \pm \sqrt{289 - 288}}{24}$$

$$= \frac{17 \pm \sqrt{1}}{24}$$

Work out the + and - parts separately to get the two solutions

$$x = \frac{17+1}{24} = \frac{18}{24}$$

$$x = \frac{17-1}{24} = \frac{16}{24}$$

$$\boldsymbol{x = \frac{3}{4}, \quad x = \frac{2}{3}}$$

Your notes

Completing the Square

Your notes

Completing the Square

What is completing the square?

- **Completing the square** is another way of writing a quadratic function
- It means rewriting $y = ax^2 + bx + c$ in the form $y = a(x + p)^2 + q$
  - The key point is that x now only occurs once in the equation
- It can be used to solve quadratic equations, sketch their graphs and to find the coordinates of the turning point

How do I complete the square?

The method used will depend on the value of the coefficient of the x^2 term in $y = ax^2 + bx + c$

- When $a = 1$
  - p is half of b
  - q is $c - p^2$

e.g. $y = x^2 + 8x - 2$

STEP 1: "$p = \frac{b}{2}$" $\quad p = \frac{8}{2} = 4$

STEP 2: "$q = c - p^2$" $\quad q = -2 - 4^2 = -18$

STEP 3: WRITE THE ANSWER!

$y = (x + 4)^2 - 18$

- When **a ≠ 1**
  - First take a factor of a out of the x^2 and x terms
  - Then continue as above

NO NEED TO INVOLVE "c"

e.g. $y = 4x^2 + 16x + 5$

STEP 1: FACTOR "a" ON RHS $y = 4[x^2 + 4x] + 5$

STEP 2: WORKING WITH $[x^2 + 4x]$ ONLY, "$p = \frac{b}{2}$"

$p = \frac{4}{2} = 2$

STEP 3: STILL WITH $[x^2 + 4x]$ ONLY, "$q = c - p^2$"

$q = 0 - 2^2 = -4$

STEP 4: EXPAND AND SIMPLIFY $y = 4[(x+2)^2 - 4] + 5$

$y = 4(x+2)^2 - 16 + 5$

$y = 4(x+2)^2 - 11$

Examiner Tips and Tricks

- Sometimes a question will explicitly use the phrase **complete the square**
- Sometimes a question will use the form $a(x + p)^2 + q$ without using the phrase completing the square

Worked Example

Write $3x^2 - 12x + k$ in the form $a(x+p)^2 + k + q$, where $a,\ p$ and q are constants to be found, and k is an unknown constant.

The form required is 'completing the square' (do not be put off by the k, it is just a constant!).

STEP 1 - Take a factor of 3 out of the x^2 and x terms - leaving k avoids awkward fractions.

$$3x^2 - 12x + k = 3(x^2 - 4x) + k$$

STEPS 2 and 3 - Complete the square on the $(x^2 - 4x)$ part only. "$p = -\frac{4}{2} = -2$" and "$q = 0 - (-2)^2 = 4$".

$$3[(x-2)^2 - 4] + k$$

STEP 4 - Expand and simplify.

$$3(x-2)^2 - 12 + k$$

$$\mathbf{3(x-2)^2 + k - 12}$$

i.e. $a = 3,\ p = -2,\ q = -12$

Solving by Completing the Square

How do I solve a quadratic equation by completing the square?

- To solve $x^2 + bx + c = 0$
  - **replace** the first two terms, $\boldsymbol{x^2 + bx}$, with $\boldsymbol{(x+p)^2 - p^2}$ where **p is half of b**
  - this is called **completing the square**
    - $x^2 + bx + c = 0$ becomes
      - $(x+p)^2 - p^2 + c = 0$ where p is half of b
  - rearrange this equation to **make x the subject** (using ±√)
- For example, solve $x^2 + 10x + 9 = 0$ by completing the square
  - $x^2 + 10x$ becomes $(x+5)^2 - 5^2$
  - so $x^2 + 10x + 9 = 0$ becomes $(x+5)^2 - 5^2 + 9 = 0$
  - make x the subject (using ±√)

Your notes

- $(x + 5)^2 - 25 + 9 = 0$
- $(x + 5)^2 = 16$
- $x + 5 = \pm\sqrt{16}$
- $x = \pm 4 - 5$
- $x = -1$ or $x = -9$
- If the equation is $ax^2 + bx + c = 0$ with a number in front of x^2, then divide both sides by a first, before completing the square

Examiner Tips and Tricks

- When making x the subject to find the solutions at the end, don't expand the squared brackets back out again!
    - Remember to use $\pm\sqrt{}$ to get **two** solutions

Worked Example

Solve $2x^2 - 8x - 24 = 0$ by completing the square.

Divide both sides by 2 to make the quadratic start with x^2

$$x^2 - 4x - 12 = 0$$

Halve the middle number, -4, to get -2
Replace the first two terms, $x^2 - 4x$, with $(x - 2)^2 - (-2)^2$

$$(x-2)^2 - (-2)^2 - 12 = 0$$

Simplify the numbers

$$(x-2)^2 - 4 - 12 = 0$$
$$(x-2)^2 - 16 = 0$$

Add 16 to both sides

$$(x-2)^2 = 16$$

Square root both sides
Include the ± sign to get two solutions

$$x - 2 = \pm\sqrt{16} = \pm 4$$

Add 2 to both sides

$$x = \pm 4 + 2$$

Work out each solution separately

x = 6 or x = -2

Your notes

Quadratic Equation Methods

Your notes

Quadratic Equation Methods

If you have to solve a quadratic equation but are not told which method to use, here is a guide as to what to do

When should I solve by factorisation?

- When the question asks to **solve by factorisation**
  - For example, part (a) Factorise $6x^2 + 7x - 3$, part (b) Solve $6x^2 + 7x - 3 = 0$
- When solving **two-term quadratic equations**
  - For example, solve $x^2 - 4x = 0$
    - ...by taking out a **common factor** of x to get $x(x - 4) = 0$
    - ...giving $x = 0$ and $x = 4$
  - For example, solve $x^2 - 9 = 0$
    - ...using the **difference of two squares** to factorise it as $(x + 3)(x - 3) = 0$
    - ...giving $x = -3$ and $x = 3$
    - (Or by rearranging to $x^2 = 9$ and using ±√ to get $x =$ $= \pm 3$)
- When possible, factorising is usually the easiest way to solve a quadratic equation
  - Even on the calculator paper, if you can spot a factorisation quickly, use this approach

When should I use the quadratic formula?

- If the coefficients (a, b and c) are **large, factorising** and **completing the square** can be **difficult** or **slow**
  - The quadratic formula lends itself to using a **calculator**
  - Some modern calculators will solve quadratic equations directly, with no need to use the formula
- Typically the quadratic formula would be used when rounding is involved
  - For example, if a question says to leave solutions correct to 2 decimal places or 3 significant figures
- However, the quadratic formula is also useful when answers need to be exact
  - The formula lends itself to surd form after simplifying some of the values within it

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- e.g. $x = \frac{-4 \pm \sqrt{4^2 - 4 \times 2 \times (-2)}}{2 \times 2} = \frac{-4 \pm \sqrt{32}}{4} = \frac{-4 \pm 4\sqrt{2}}{4} = -1 \pm \sqrt{2}$
- **If in doubt**, use the quadratic formula - it always works

When should I solve by completing the square?

- A question may direct you to solve by completing the square
  - e.g. Part (a) says to **complete the square** and part (b) says 'hence' or 'use part (a)' to solve ...
- Completing the square may have already happened for other reasons
  - e.g. Completing the square allows the coordinates of the turning point on a quadratic graph to be found easily
  - If this has been done in an earlier part of a question, use it to solve the quadratic equation

Examiner Tips and Tricks

- Calculators can solve quadratic equations
  - Double check you've entered the equation correctly, in the correct format
  - Use this feature to check your answers where possible
  - If the solutions on your calculator are whole numbers or fractions (with no square roots), this means the quadratic equation does factorise

Worked Example

a) Solve $x^2 - 7x + 2 = 0$, giving your answers correct to 2 decimal places

"Correct to 2 decimal places" suggests using the quadratic formula and/or a calculator
For accuracy, it is a good idea to use both - use the formula and calculator as normal first
Then use the quadratic solver feature to check your solutions

Substitute *a* = 1, *b* = -7 and c = 2 into the formula, putting brackets around any negative numbers

$$x = \frac{-(-7) \pm \sqrt{(-7)^2 - 4 \times 1 \times 2}}{2 \times 1}$$

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Use a calculator to find each solution

x = 6.70156... or 0.2984...

Round your final answers to 2 decimal places

x = 6.70 or x = 0.30

If your calculator has a quadratic equation solver, use it to check your answers

(b) Solve $16x^2 - 82x + 45 = 0$

Method 1

The coefficients are large and so the factorisation, even if possible, is hard to spot

Therefore, one method to use is the quadratic formula - it always works!

The solution below is the manual way to use a calculator, but as above, if your calculator has a quadratic solver feature, you may use that

Substitute a = 16, b = -82 and c = 45 into the formula, putting brackets around any negative numbers

$$x = \frac{-(-82) \pm \sqrt{(-82)^2 - 4 \times 16 \times 45}}{2 \times 16}$$

Use a calculator to find each solution

$x = \frac{9}{2}$ or $x = \frac{5}{8}$

Method 2

If you do persevere with the factorisation then use that method instead

$$16x^2 - 82x + 45 = (2x - 9)(8x - 5) = 0$$

Set the first bracket equal to zero

$$2x - 9 = 0$$

Add 9 to both sides then divide by 2

Your notes

$$2x = 9$$
$$x = \frac{9}{2}$$

Set the second bracket equal to zero

$$8x - 5 = 0$$

Add 5 to both sides then divide by 8

$$8x = 5$$
$$x = \frac{5}{8}$$

$$\mathbf{x = \frac{9}{2} \text{ or } x = \frac{5}{8}}$$

(c) By writing $x^2 + 6x + 5$ in the form $(x + p)^2 + q$, solve $x^2 + 6x + 5 = 0$

Notice this question does not use the phrase 'completing the square' but shows the form of it instead
Find p (by halving the middle number)

$$p = \frac{6}{2} = 3$$

Write $x^2 + 6x$ as $(x + p)^2 - p^2$

$$x^2 + 6x = (x + 3)^2 - 3^2$$
$$= (x + 3)^2 - 9$$

Replace $x^2 + 6x$ with $(x + 3)^2 - 9$ in the equation

$$(x+3)^2 - 9 + 5 = 0$$
$$(x+3)^2 - 4 = 0$$

Make x the subject of the equation (start by adding 4 to both sides)

$$(x+3)^2 = 4$$

Take square roots of both sides (include a ± sign to get both solutions)

$$x + 3 = \pm\sqrt{4} = \pm 2$$

Subtract 3 from both sides

$$x = \pm 2 - 3$$

Find each solution separately using + first, then - second

$x = -5, x = -1$

Even though the quadratic factorises to $(x + 5)(x + 1)$, this is not the method asked for in the question

Hidden Quadratic Equations

How do I spot a hidden quadratic equation?

- **Hidden** quadratics have the same **structure** as quadratic equations
  - $a(\text{something})^2 + b(\text{something}) + c = 0$
- Here are some hidden quadratics based on $x^2 - 3x - 4 = 0$:
  - $x^4 - 3x^2 - 4 = 0$ (a quadratic in x^2)
  - $x^{16} - 3x^8 - 4 = 0$ (a quadratic in x^8)
  - $x - 3\sqrt{x} - 4 = 0$ (a quadratic in $\sqrt{x}$ because $(\sqrt{x})^2$ is x)
  - $x^{\frac{2}{3}} - 3x^{\frac{1}{3}} - 4 = 0$ (a quadratic in $x^{\frac{1}{3}}$ because $\left(x^{\frac{1}{3}}\right)^2 = x^{\frac{2}{3}}$)

Your notes

Your notes

- Sometimes, a **change of base** helps to spot a hidden quadratic
  - e.g. the first term in $4^x - 3 \times 2^x - 4 = 0$ can be written $4^x = (2^2)^x = 2^{2x} = (2^x)^2$
    - $(2^x)^2 - 3 \times 2^x - 4 = 0$ is a quadratic in 2^x
- **Trigonometric** equations can also be in the form of a quadratic
  - e.g. $3\tan^2 3x + 4\tan 3x - 6 = 0$ is a quadratic in $\tan 3x$

How do I solve a hidden quadratic equation?

- You can solve $a(...)^2 + b(...) + c = 0$ with a **substitution**
  - **Substitute** "$\boldsymbol{u}$ = ..." and rewrite the equation in terms of ***u* only**
    - $au^2 + bu + c = 0$
  - **Solve** this **easier** quadratic equation in u to get $u = p$ and $u = q$
  - **Replace** the u's with their substitution to get two equations
    - "... = p" and "... = q"
  - **Solve** these **two** separate equations to find all the solutions
    - These equations might have multiple solutions or none at all!
- e.g. to solve $x^4 - 3x^2 - 4 = 0$
  - Substitute $u = x^2$ to get $u^2 - 3u - 4 = 0$
  - The solutions are $u = 4$ or $u = -1$,
  - Rewrite in terms of x:
    - $x^2 = 4$ or $x^2 = -1$,
  - Solve to give $x = -2$ or $x = 2$ (no solutions from $x^2 = -1$ as you can't square-root a negative)

Examiner Tips and Tricks

- While the substitution method is not compulsory, beware of skipping steps
  - e.g. it is incorrect to "jump" from the solutions of $x^2 - 3x - 4 = 0$ to the solutions of $(x + 5)^2 - 3(x + 5) - 4 = 0$ by "adding 5 to them"

- the substitution method shows you end up subtracting 5

Your notes

Worked Example

(a) Solve $x^8 - 17x^4 + 16 = 0$

This is a quadratic in x^4 so let $u = x^4$

$$u^2 - 17u + 16 = 0$$

Solve this simpler quadratic equation, for example by factorisation

$$(u - 16)(u - 1) = 0$$

Write out the u solutions

$$u = 16 \quad \text{or} \quad u = 1$$

Replace u with x^4

$$x^4 = 16 \quad \text{or} \quad x^4 = 1$$

Solve these separate equations (remember an even power gives two solutions)

$$x = \pm 2 \quad \text{or} \quad x = \pm 1$$

Write your solutions out (it's good practice to write them in numerical order)

$$\boldsymbol{x = -2,\ -1,\ 1 \text{ or } 2}$$

(b) Solve $x - \sqrt{x} - 6 = 0$

This is a quadratic in $\sqrt{x}$ so let $u = \sqrt{x}$

$$u^2 - u - 6 = 0$$

Solve this simpler quadratic equation, for example by factorisation

$$(u - 3)(u + 2) = 0$$
$$u = 3 \text{ or } u = -2$$

Replace u with $\sqrt{x}$ and solve

$$\sqrt{x} = 3 \Rightarrow x = 9$$
$$\sqrt{x} = -2 \text{ has no solutions as } \sqrt{x} \geq 0$$

You can check your solutions by substituting them back into the equation If you put $x = 4$ as a solution by mistake then substituting will spot this error

$$9 - \sqrt{9} - 6 = 0$$
$$4 - \sqrt{4} - 6 = -4 \neq 0$$

$$\boldsymbol{x = 9}$$

Your notes

Discriminants

Your notes

Discriminants

What is a discriminant?

- The **discriminant** is the part of the quadratic formula that is **under the square root** sign
- It is $b^2 - 4ac$
    - The **quadratic formula,** in full, is

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

where the quadratic equation is written in the form

$$ax^2 + bx + c = 0$$

- It is sometimes denoted by the Greek letter Δ (capital delta)

Worked Example

Find, in terms of the constant k, the discriminant of the quadratic equation $3x^2 + 2kx - k = kx^2 - 4kx + 2$.

First write the quadratic equation in the form $ax^2 + bx + c = 0$.

$$\begin{aligned} 3x^2 - kx^2 + 2kx + 4kx - k - 2 &= 0 \\ (3 - k)x^2 + 6kx - (k + 2) &= 0 \end{aligned}$$

It can be easier/clearer to pick out $a,$ b and c first, before finding the discriminant.

$$\begin{aligned} a &= 3 - k \\ b &= 6k \\ c &= -(k + 2) \end{aligned}$$

The discriminant is $b^2 - 4ac$.

Your notes

$$\Delta = (6k)^2 - 4 \times (3-k) \times -(k+2)$$
$$\Delta = 36k^2 + 4(6 + k - k^2)$$

$$\boldsymbol{\Delta = 32k^2 + 4k + 24}$$

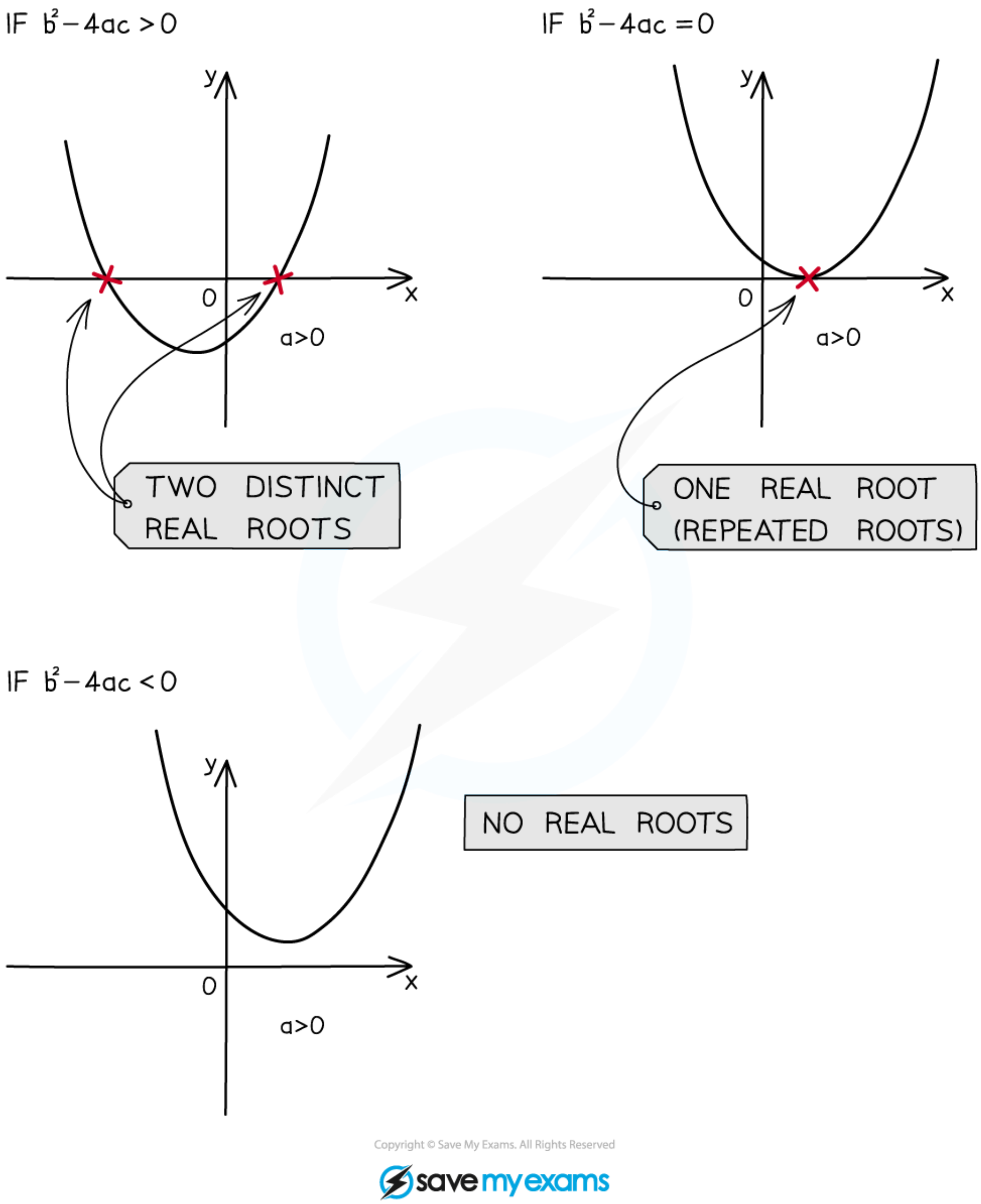
Applications of Discriminant

How does the value of the discriminant affect roots?

- There are three options for the outcome of the discriminant:
  - If $b^2 - 4ac > 0$ the square root part of the quadratic formula can be calculated leading to two solutions (values of x)
    - i.e. **two** different **real roots**
  - If $b^2 - 4ac = 0$ the square root part of the quadratic formula will be zero leading to one solution
    - i.e. **one repeated** root or **two equal roots**
  - If $b^2 - 4ac < 0$ the square root part of the quadratic formula cannot be calculated leading to no solutions
    - i.e. **no (real) roots**

How do I sketch quadratic graphs using the discriminant?

- If $b^2 - 4ac > 0$ the quadratic equation has **two** different real **roots**
  - The graph of the quadratic will **intersect** the x-axis **twice** (at the roots)
- If $b^2 - 4ac = 0$ the quadratic equation has two equal roots **(one root)**
  - The graph of the quadratic will **intersect** (touch) the x-axis **once** (at the root)
- If $b^2 - 4ac < 0$ the quadratic equation has **no** (real) **roots**
  - The graph of the quadratic will **not intercept** the x-axis



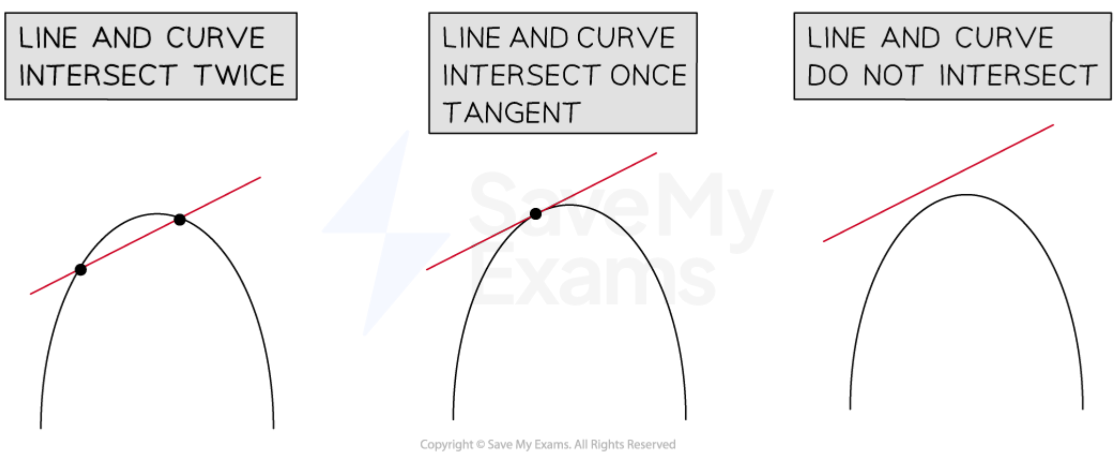
How do I use the discriminant to find the number of intersections between a line and a curve?

- For the graphs of two functions, $y = \mathrm{f}(x)$ and $y = \mathrm{g}(x)$ where

- $\mathrm{f}(x)$ is **quadratic**
- $\mathrm{g}(x)$ is **linear**

the number of intersections between the graphs can be found using the discriminant.

- **STEP 1**
  - Set $\mathrm{f}(x) = \mathrm{g}(x)$
- **STEP 2**
  - Rearrange into the form $\mathrm{h}(x) = 0$ such that $\mathrm{h}(x)$ is in the quadratic form $ax^2 + bx + c = 0$
- **STEP 3**
  - Find the **discriminant** and thus determine the **number** of **intersections** between the graphs of $y = \mathrm{f}(x)$ and $y = \mathrm{g}(x)$
    - if $b^2 - 4ac > 0$ (two real roots) the graphs **intersect twice**
    - if $b^2 - 4ac = 0$ (equal roots) the graphs **intersect once**
    - this means the line ($\mathrm{g}(x)$) is a tangent to the curve ($\mathrm{f}(x)$)
    - if $b^2 - 4ac < 0$ (no real roots) the graphs **do not intersect**



Your notes

Worked Example

Show that the line with equation $y = 3x - 7$ is tangent to the curve with equation $y = (x+3)(x-2)$.

STEP 1 - set the equations of the line and curve equal to each other.

$$3x - 7 = (x+3)(x-2)$$

STEP 2 - rearrange to quadratic form.

$$3x - 7 = x^2 + x - 6$$
$$x^2 - 2x + 1 = 0$$

STEP 3 - find the discriminant and interpret its value.

$$\Delta = (-2)^2 - 4(1)(1) = 0$$

Since the discriminant is zero, the line and the curve intersect at one point only. Therefore the line is a tangent to the curve.

Your notes

Quadratic Graphs

Your notes

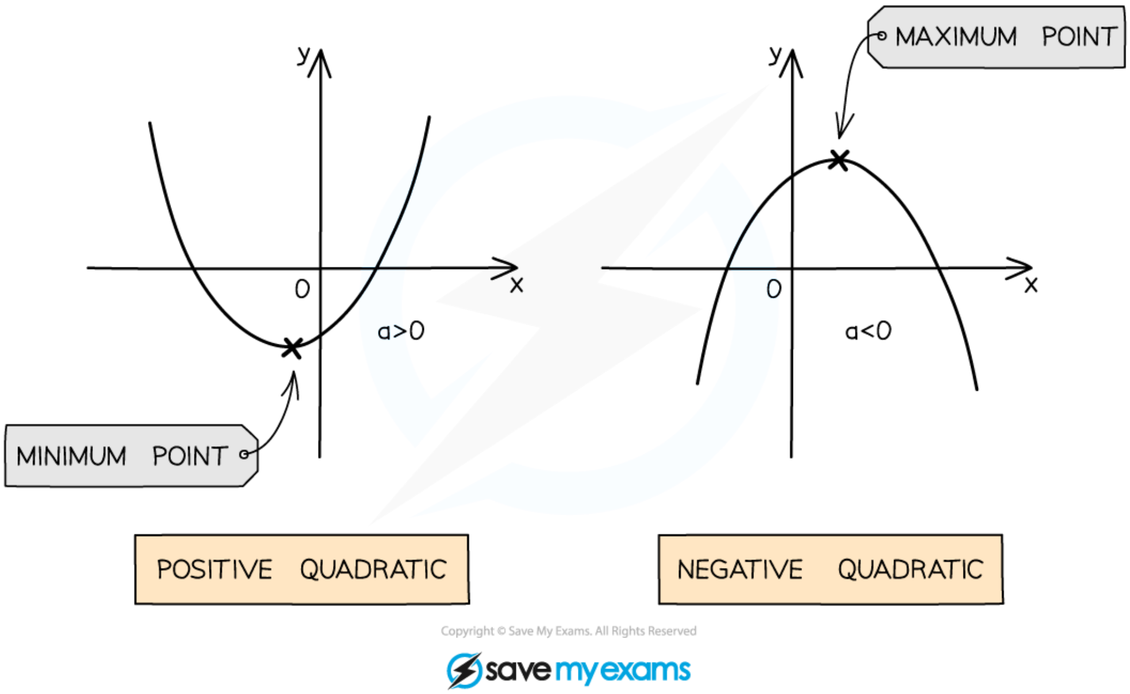
Quadratic Graphs

A quadratic is a function of the form $y = ax^2 + bx + c$ where a is not zero They are a very common type of function in mathematics, so it is important to know their key features

What does a quadratic graph look like?

- The shape made by a quadratic graph is known as a **parabola**
- The parabola shape of a quadratic graph can either look like a "∪-shape" or a "∩-shape"
  - A quadratic with a **positive** coefficient of x^2 will be a ∪-shape
  - A quadratic with a **negative coefficient** of x^2 will be a ∩-shape
- A quadratic will always cross the y-axis
- A quadratic may cross the x-axis twice, once, or not at all
  - The points where the graph crosses the x-axis are called the **roots**
- If the quadratic is a ∪-shape, it has a **minimum point** (the bottom of the ∪)
- If the quadratic is a ∩-shape, it has a **maximum point** (the top of the ∩)
- Minimum and maximum points are both examples of turning points

Your notes



How do I sketch a quadratic graph?

- We could create a table of values for the function and then plot it accurately
  - However we often only require a sketch to be drawn, showing just the key features
- The key features needed to be able to sketch a quadratic graph are
  - the overall shape
    - ∪-shape graphs occur when $a > 0$ (positive quadratic)
    - ∩-shape graphs occur when $a < 0$ (negative quadratic)
  - the x-intercept(s), these are also known as the **roots** (there may be none!)
    - roots are found by setting the quadratic function (or y) equal to zero
    - i.e. solve $ax^2 + bx + c = 0$
    - if there are **no** (real) **solutions** (i.e. no roots), the graph does **not intersect** the x-axis
      - the discriminant can be used to determine whether a quadratic function has 0, 1 or 2 roots

Your notes

- the y-intercept
  - this is found by setting $x = 0$ in the quadratic function
  - so for $ax^2 + bx + c$ the coordinates of the y-intercept will be $(0,\ c)$
- the **minimum** or **maximum** point (turning point)
  - sometimes a rough idea of where this should lie is enough
  - sometimes the specific coordinates of the turning point will be needed
  - when required the coordinates of the turning point can be found by either **completing the square** or **differentiation**
    - in cases where the quadratic has just **one root,** the graph will *touch* (rather than *cross)* the x-axis and so this will be the turning point

Worked Example

a) Sketch the graph of $y = x^2 - 5x + 6$, labelling any intercepts with the coordinate axes.

It is a positive quadratic, so will be a $\cup$ -shape

The '$+c$' at the end is the y-intercept ($y = c$ when $x = 0$), so the graph crosses the y-axis at (0,6)

Factorise

$$y = (x-2)(x-3)$$

Solve $y = 0$

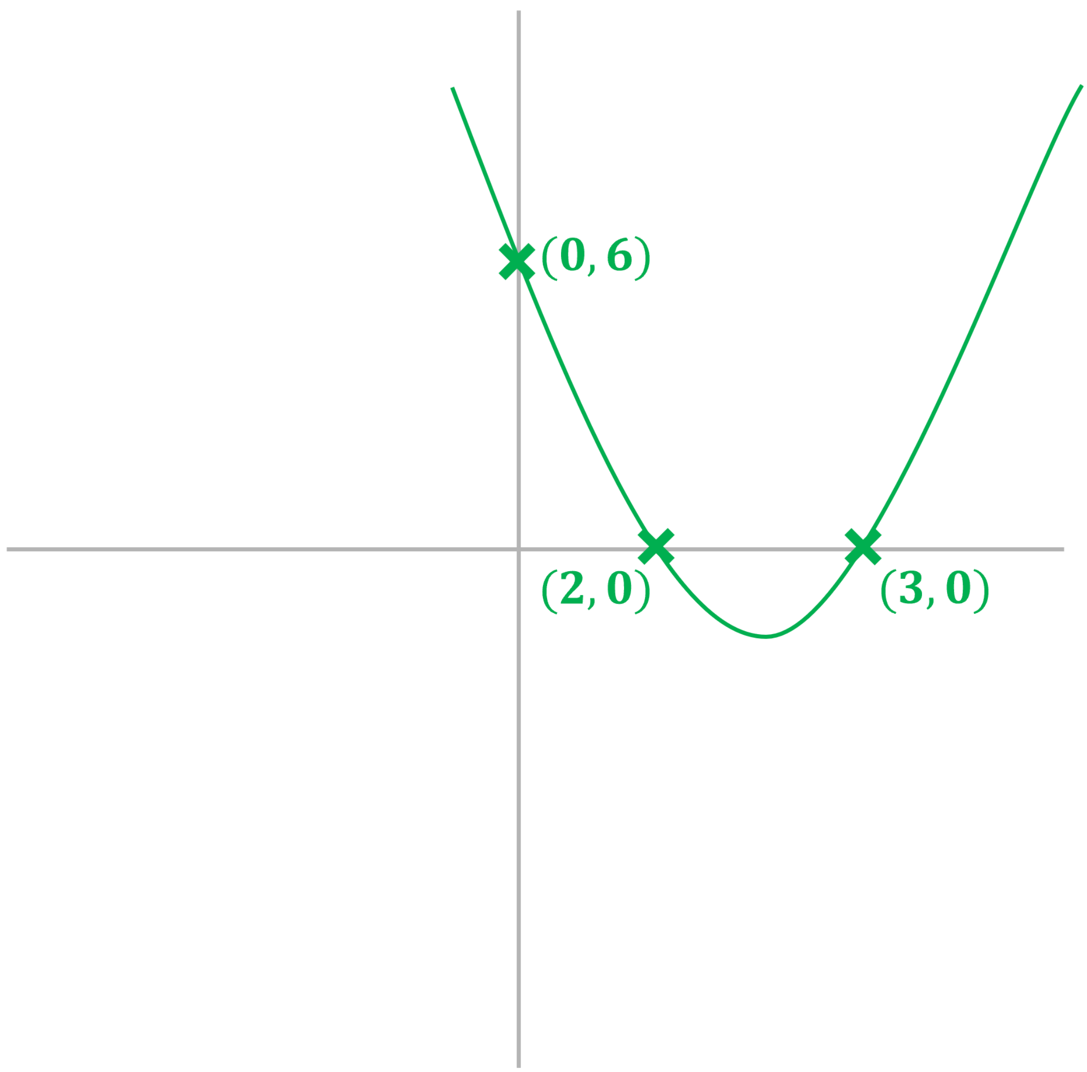
$$(x-2)(x-3) = 0$$

$$x = 2 \text{ or } x = 3$$

So the roots of the graph are

(2,0) and (3,0)

Your notes



b) Sketch the graph of $y = x^2 - 6x + 13$, labelling any intercepts with the coordinate axes.

It is a positive quadratic, so will be a U -shape

The '$+c$' at the end is the y-intercept, so this graph crosses the y-axis at

(0,13)

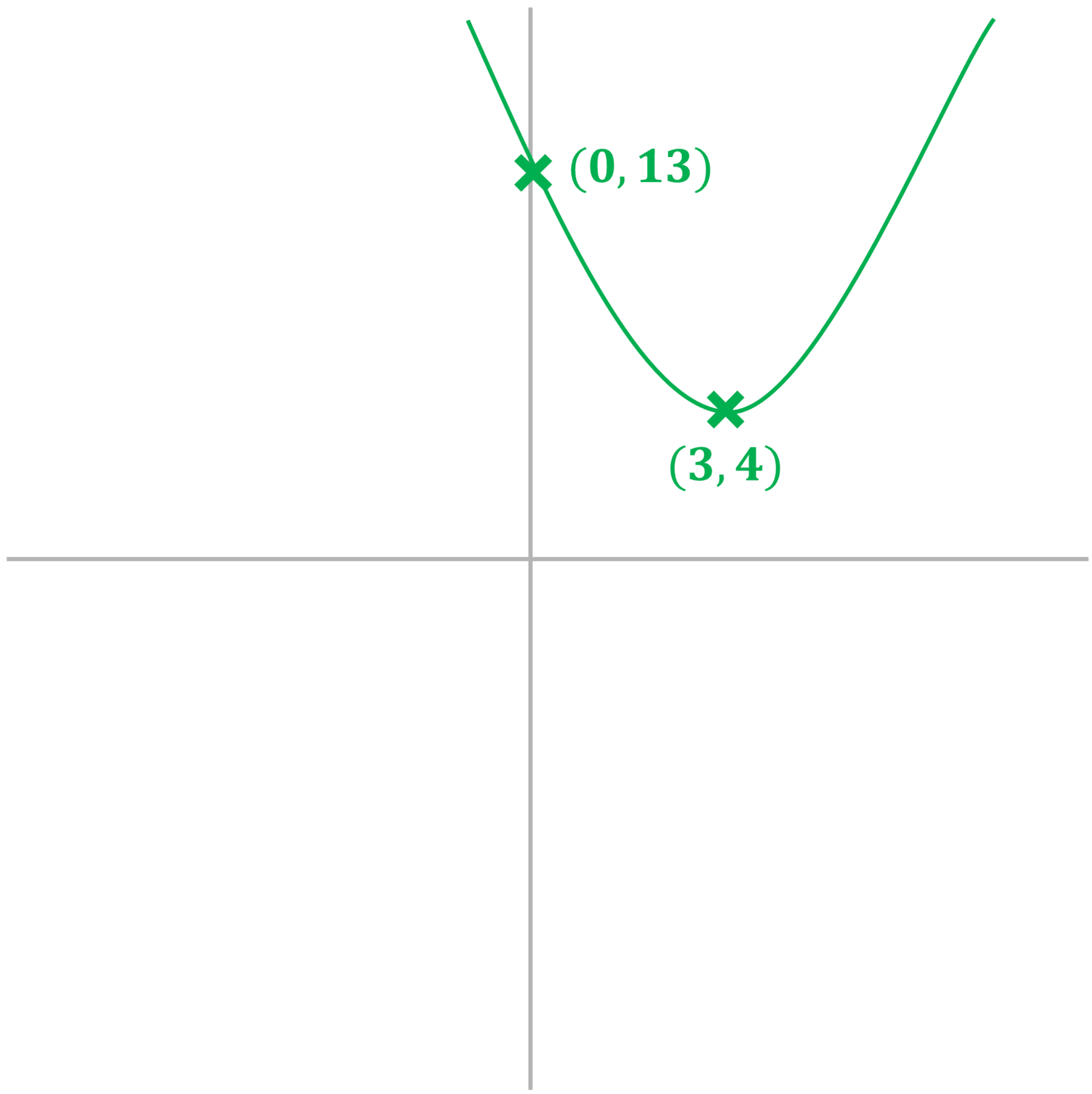
The discriminant of the quadratic is '$b^2 - 4ac$'

$$(-6)^2 - 4(1)(13) = 36 - 52 = -16$$

Your notes

As the discriminant is negative, there are no (real) roots and the graph does not intersect the x-axis

(Note we have included the coordinates of the turning point, $(3, 4)$ to help you visualise the graph, but there was no requirement from the question to do this - on a sketch like this, the turning point should be in the correct quadrant)



c) Sketch the graph of $y = -x^2 - 4x - 4$, labelling any intercepts with the coordinate axes and the turning point.

It is a negative quadratic, so will be an $\cap$ -shape

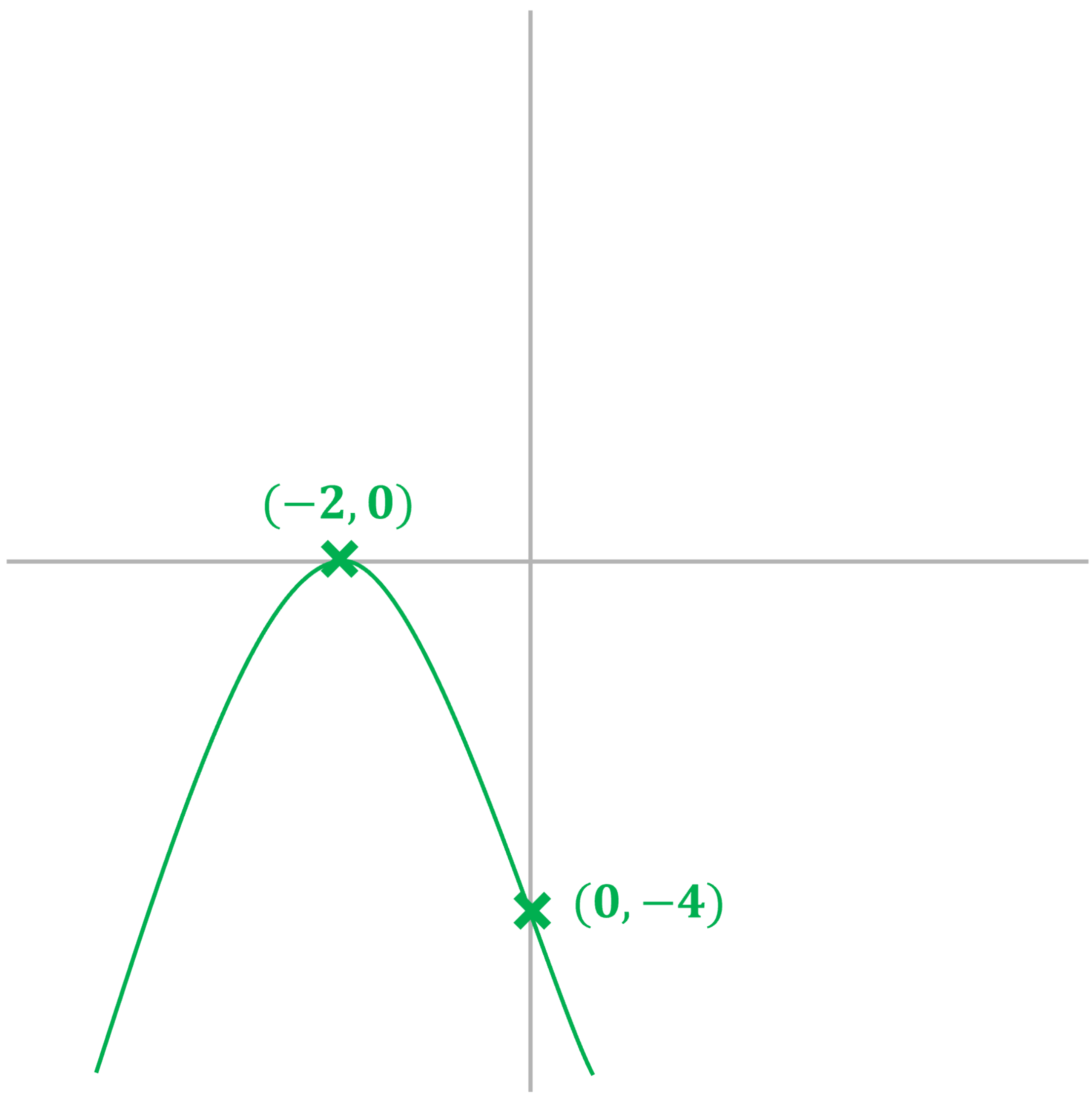
Your notes

The '$+c$' at the end is the y-intercept, so this graph crosses the y-axis at (0, -4)

Factorising

$$-x^2 - 4x - 4 = -(x^2 + 4x + 4) = -(x+2)^2$$

This shows that there is only one root and the graph will touch the x-axis at the point (-2, 0) This point will also be the turning point - and as this is a negative quadratic - will be a maximum point



Sketching Graphs by Completing the Square

How does completing the square help me sketch graphs?

- Completing the square can quickly tell us the coordinates of the turning point on a quadratic graph
- This is based on the fact that a **squared term** (e.g. $(x+1)^2$) cannot be **negative**
- **STEP 1**
  Complete the square - rewrite $ax^2 + bx + c = 0$ in the form $a(x+p)^2 + q$
- **STEP 2**
  Deduce the $\boldsymbol{x}$**-coordinate** of the turning point
- $(x+p)^2 \geq 0$ for all values of x
  - Therefore it's minimum value is 0, and this occurs when $x = -p$

The x-coordinate is $-p$

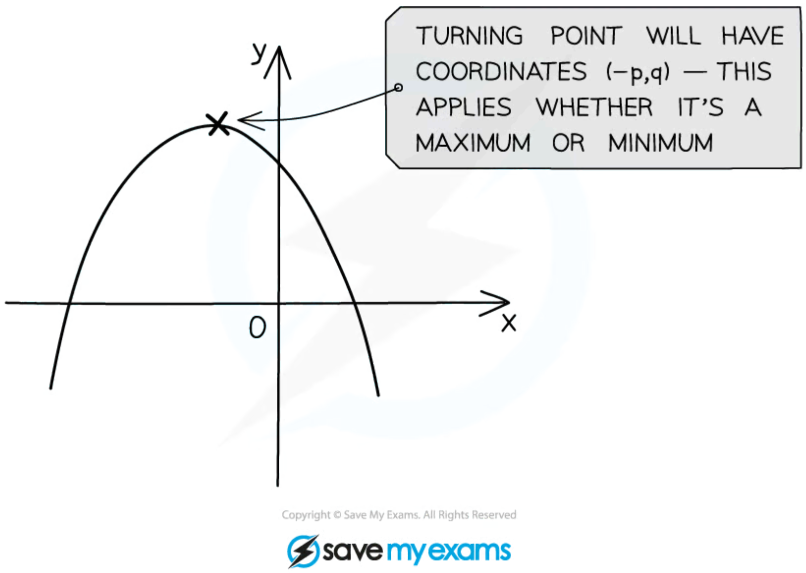
- **STEP 3**
  Deduce the $\boldsymbol{y}$**-coordinate** of the turning point
- $a(x+p)^2 = 0$
  - Therefore $y = q$

The y-coordinate is q

- **STEP 4**
  The **turning point** has coordinates $(-p,\ q)$ This can be considered when sketching the graph of the quadratic function
- Note that the turning point could be a **maximum** or **minimum** point - this will depend on the value of a
  - a is the **coefficient** of the x^2 term
  - If a is **positive,** the graph is **U - shaped** and will have a **minimum** point
  - If a is **negative,** the graph is **∩ - shaped** and will have a **maximum** point

Your notes

Your notes



How do I use the graph of a quadratic function to find its range?

- The **range** of a quadratic function will be shown on its graph by the values y takes
  - i.e. the **turning point** from a quadratic graph will determine its **range**
- For the quadratic function $\mathrm{f}(x)$ whose graph has a **minimum** point $\left(x_{min}, y_{min}\right)$
  - the range of the $\mathrm{f}(x)$ will be $\mathrm{f} \geq y_{min}$
- For the quadratic function $\mathrm{f}(x)$ whose graph has a **maximum** point $\left(x_{max}, y_{max}\right)$
  - the range of $\mathrm{f}(x)$ will be $\mathrm{f} \leq y_{max}$
- If there any restrictions on the domain of $\mathrm{f}(x)$ then they could affect the range of $\mathrm{f}(x)$

Worked Example

Sketch the graph of $y = \mathrm{f}(x)$ where $\mathrm{f}(x) = 2x^2 - 4x - 6$, giving the coordinates of the turning point, and any points where the graph intercepts the coordinate axes. Use your graph to write down the range of $\mathrm{f}(x)$.

Your notes

STEP 1 - Complete the square.

$$\begin{aligned} f(x) &= 2(x^2 - 2x) - 6 \\ &= 2[(x-1)^2 - 1] - 6 \\ &= 2(x-1)^2 - 8 \end{aligned}$$

STEP 2 - Deduce the x-coordinate.

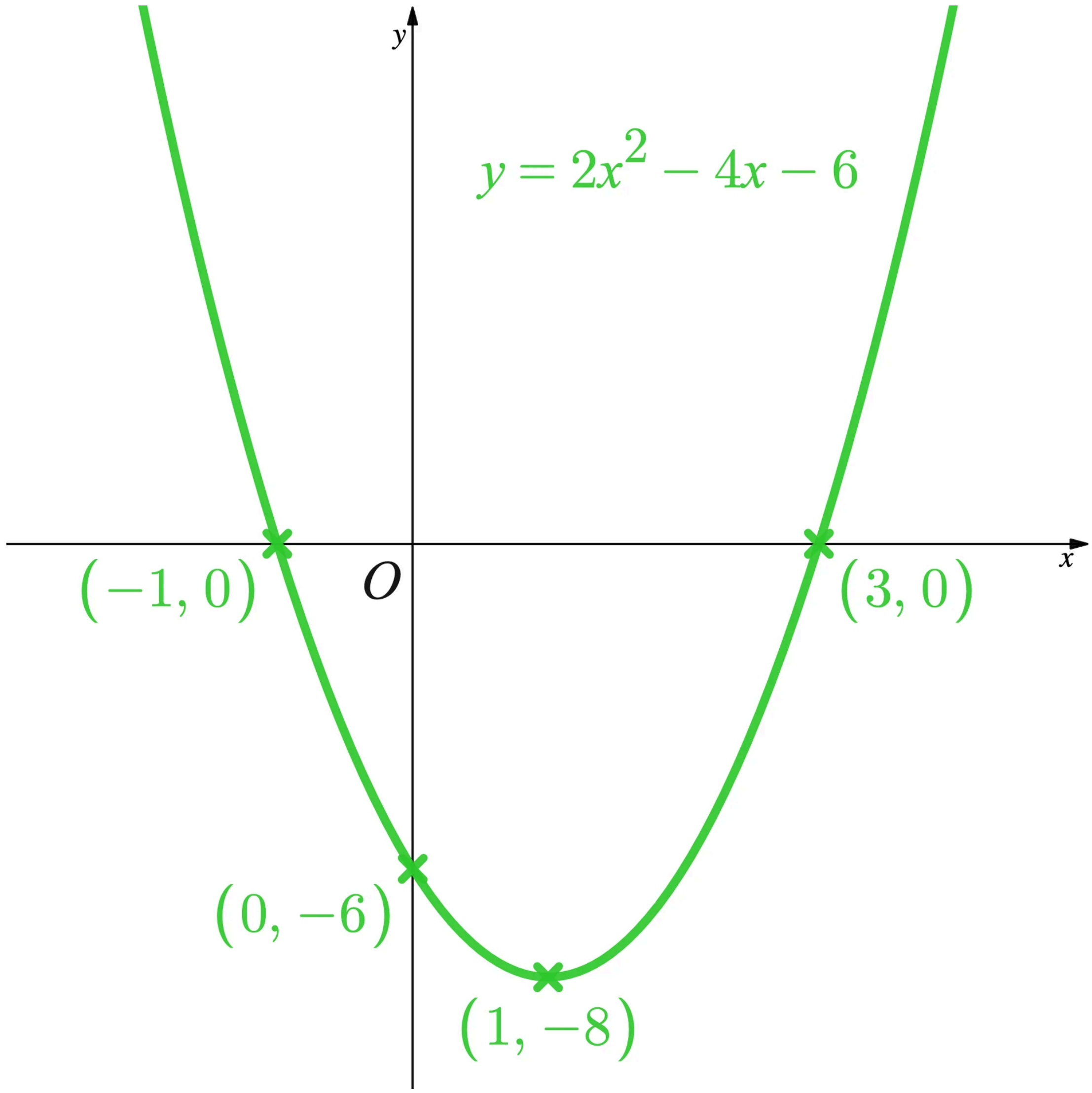
$$x = -(-1) = 1$$

STEP 3 - Deduce the y-coordinate.

$$y = -8$$

STEP 4 - Label the turning point when sketching the graph of the quadratic function.

Your notes



The graph has a minimum point so the range will be **greater than or equal to** the y-coordinate of this point.

The range of $\mathbf{f(x)}$ is $\mathbf{f > -8}$

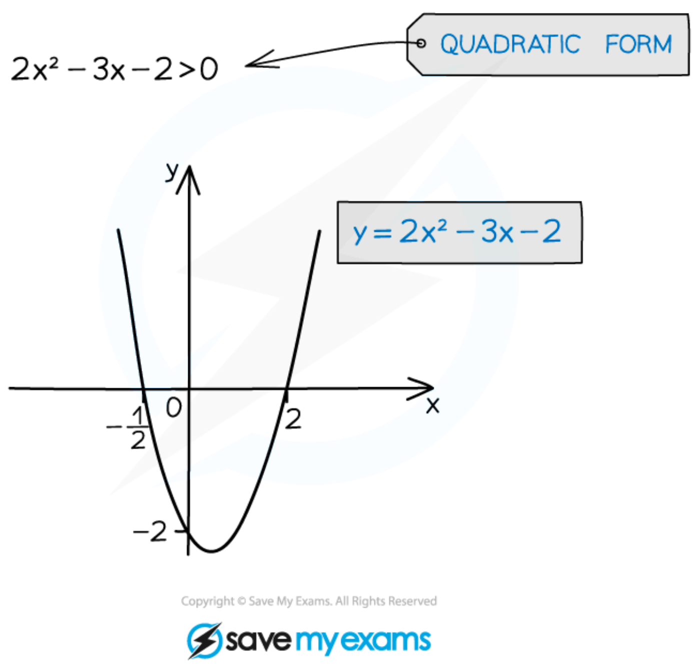
Your notes

Quadratic Inequalities

Quadratic Inequalities

What are quadratic inequalities?

- They are similar to quadratic equations with the "=" replaced by one of <, >, ≤ or ≥
  - Just like equations such inequalities should be in a form such that 0 is on one side of the inequality
    - e.g. $ax^2 + bx + c \leq 0$
- **Sketching** a **quadratic graph** is essential to finding the correct solution(s)
  - Some modern calculators may be able to solve quadratic inequalities directly
    - You could use this to check your answer

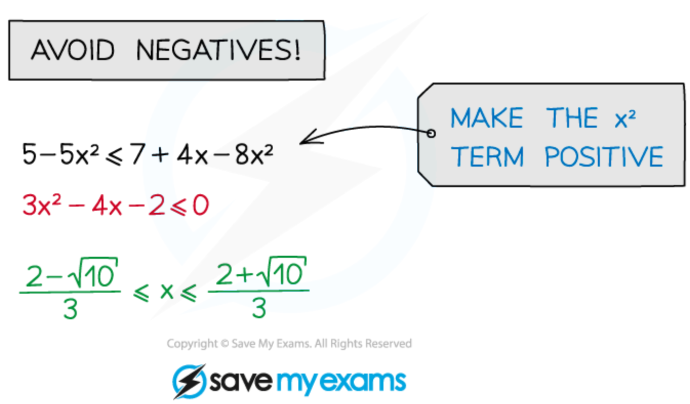


How do I solve quadratic inequalities?

Your notes

- **STEP 1**
Rearrange the inequality into quadratic form with a **positive squared term** $ax^2 + bx + c > 0$ (>, <, ≤ or ≥)
- **STEP 2**
Find the **roots** of the quadratic equation **Solve** $ax^2 + bx + c = 0$ to get x_1 and x_2 where $x_1 \leq x_2$
- **STEP 3**
Sketch the graph of the quadratic and **label the roots** As Step 1 makes the x -squared term positive it will be $\cup$ -shaped
- **STEP 4**
Identify the **region** that satisfies the inequality For $ax^2 + bx + c > 0$ you want the region **above** the *x*-axis - the solution will be $x < x_1$ **or** $x > x_2$ For $ax^2 + bx + c < 0$ you want the region below the *x*-axis - the solution will be $x_1 < x < x_2$
- Be careful:
  - **avoid** multiplying or dividing by a negative number

    if unavoidable, "**flip**" the inequality sign so **<** → **>**, **≥** → **≤**, etc
  - **do** rearrange to make the x^2 term positive



Quadratic inequalities and the discriminant

- The **discriminant** of the quadratic function $ax^2 + bx + c$ is $b^2 - 4ac$
- It's value indicates the number of (real) **roots** the quadratic function has
  - if $b^2 - 4ac > 0$ there are **two roots**

Your notes

- if $b^2 - 4ac = 0$ there is **one root** (repeated)
- if $b^2 - 4ac < 0$ there are **no roots**

- The firsts and last of these are **quadratic inequalities**
- Some questions will require you to use the discriminant to set up and solve a quadratic inequality
  - For example: Find the values of k such that the equation $(k+1)x^2 - 4x + (k-2) = 0$ has no real roots
    - Using the **discriminant,** and for **no real roots**, $(-4)^2 - 4(k+1)(k-2) < 0$
    - Using the approach above, this leads to the **quadratic inequality** in k, $k^2 - k - 6 > 0$
    - And using the method above, including **sketching a graph**, leads to the solutions $k < -2$ and $k > 3$

Examiner Tips and Tricks

- Some calculators will solve quadratic inequalities directly and just give you the answer
  - Beware!
    - make sure you have typed the inequality in correctly
    - the calculator may not display the answer in a conventional way
      - e.g $x_1 < x < x_2$ may be shown as $x_2 > x > x_1$ Both are mathematically correct but the first way is how it would normally be written
    - these questions could crop up on the non-calculator exam paper

Worked Example

Your notes

Find the set of values for which $3x^2 + 2x - 6 > x^2 + 4x - 2$.

$3x^2 + 2x - 6 - x^2 - 4x + 2 > 0$

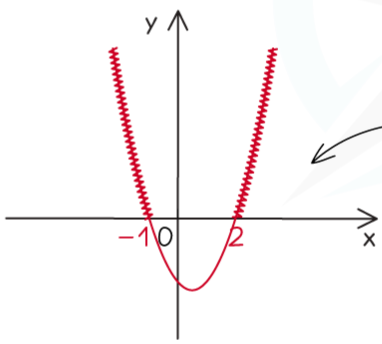
$2x^2 - 2x - 4 > 0$

REARRANGE TO QUADRATIC FORM

$2(x^2 - x - 2) > 0$

$x^2 - x - 2 > 0$

$(x - 2)(x + 1) > 0$



CRUCIAL: SKETCH THE GRAPH TO SEE WHERE THE SOLUTIONS ARE

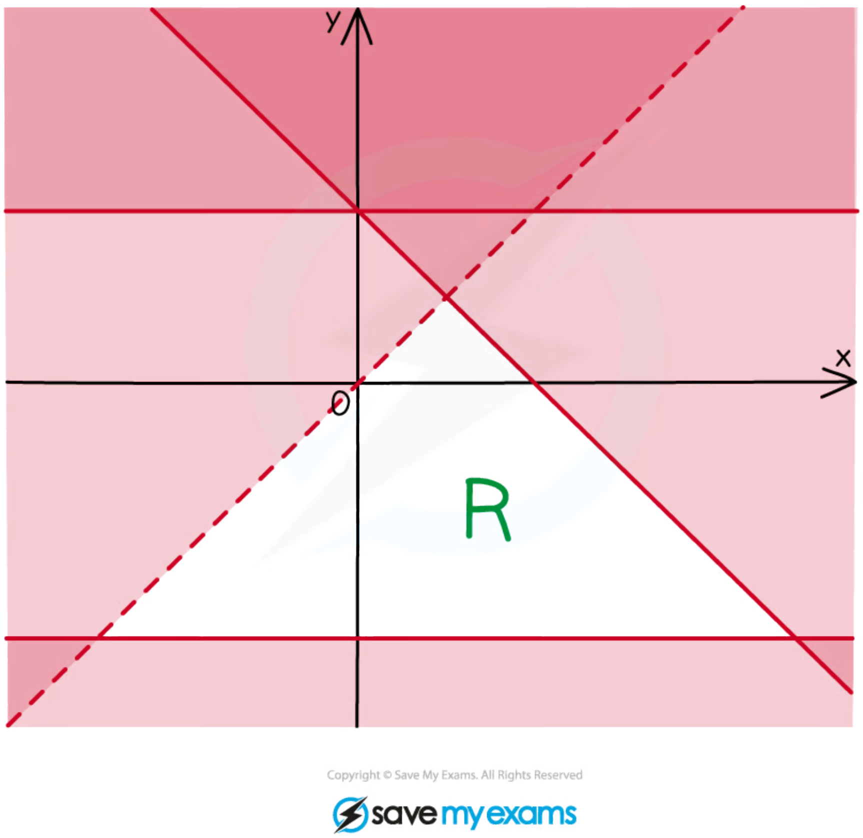
$x < -1$ OR $x > 2$

Inequalities on Graphs

What are inequalities on graphs?

- Inequalities can be represented on graphs by **shaded regions** and **dotted** or **solid** lines
- These inequalities have two variables, **x** and **y**
- Several inequalities are used at once
- The solution is an **area** on a graph (often called a **region** and labelled **R**)
- The inequalities can be **linear** or **quadratic**

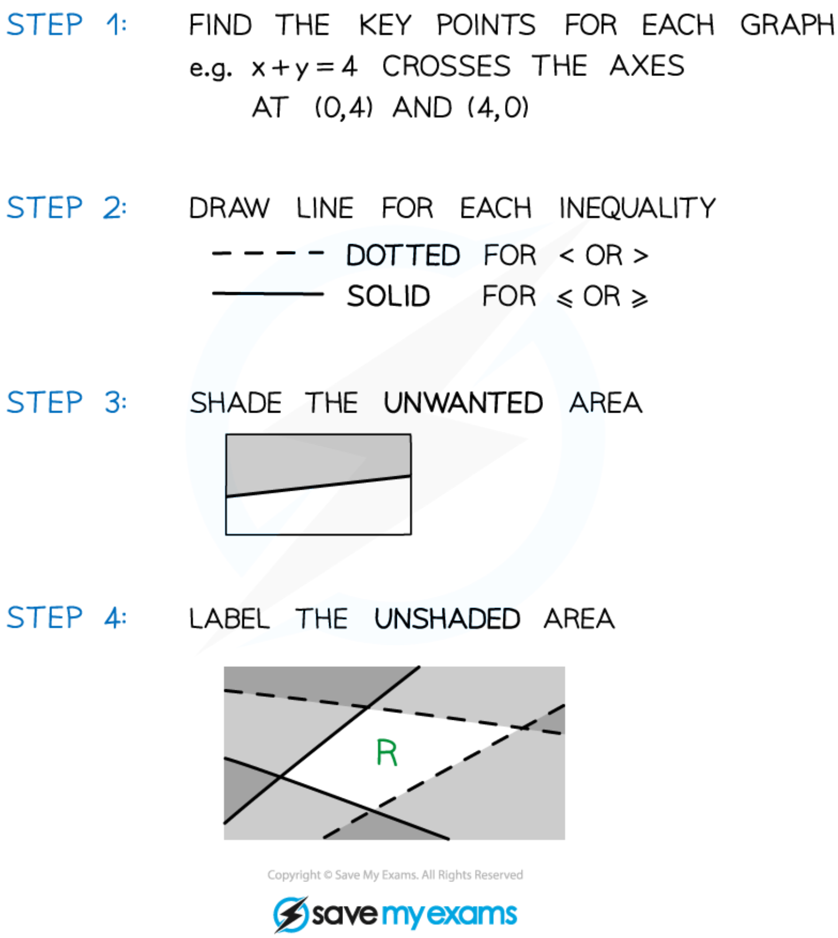
Your notes



How do I draw inequalities on a graph?

- Sketch each line or curve
  - If the inequality is **strict (< or >)** then use a **dotted line**
  - If the inequality is **weak (≤ or ≥)** then use a **solid line**
- Decide **which side** of the line **satisfies** the inequality
  - If unsure, choose a coordinate on one side and **test** it in the inequality
    - The origin is an easy point to use
  - If it satisfies the inequality then that whole side of the line satisfies the inequality
    - For example: (0,0) satisfies the inequality $y < x^2 + 1$ so you want the side of the curve that contains the origin

Your notes



Examiner Tips and Tricks

- **Recognise** this type of inequality by the use of **two** variables
- You may have to **deduce** the inequalities from a **given** graph
- Pay careful attention to which **region** you are **asked to shade**
    - Sometimes the exam could ask you to shade the region that **satisfies the inequalities this means** you should **shade** the region that is **wanted**.

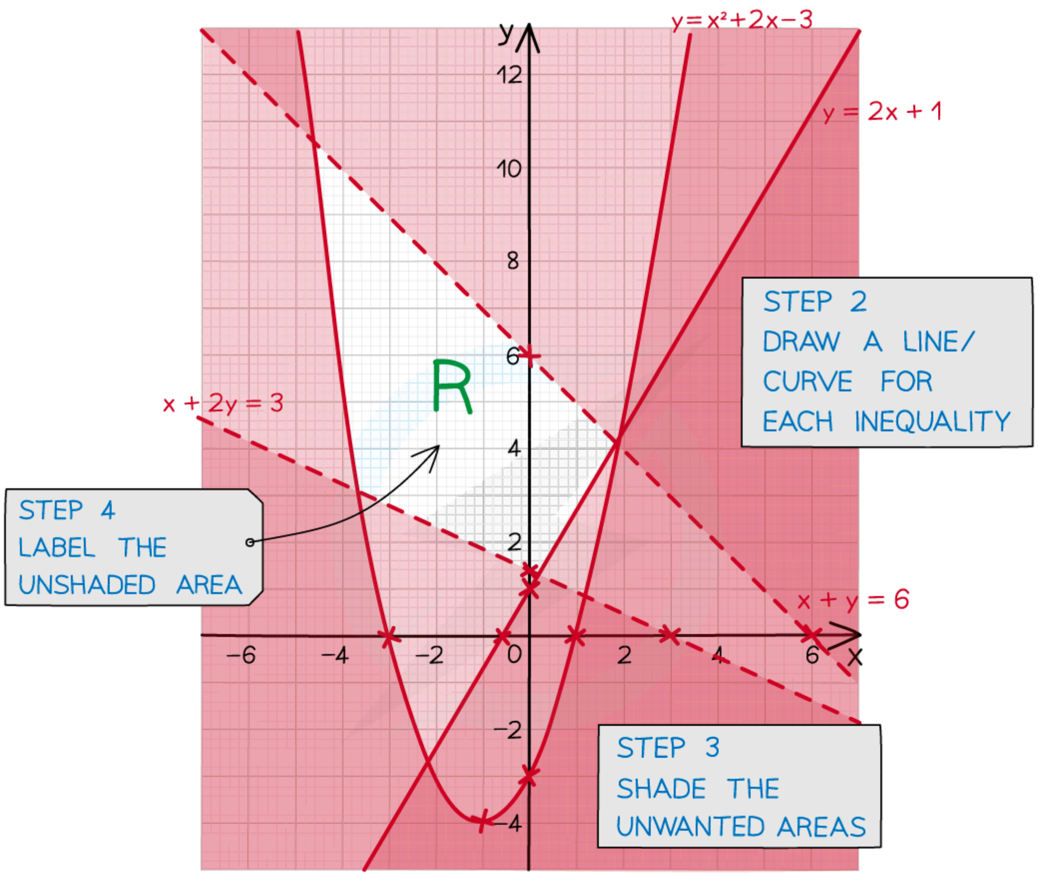
Worked Example

Your notes

Your notes

On the graph below show the region bounded by the following inequalities:

$$y \geq x^2 + 2x - 3 \qquad x + y < 6$$
$$y \geq 2x + 1 \qquad x + 2y > 3$$



TURNING POINT

$y = x^2 + 2x - 3 = (x+3)(x-1)$

$(0,-3)$ $(-3,0)$ $(1,0)$ $(-1,-4)$ "$\geq$ SOLID"

$x + y = 6$

$(0,6)$ $(6,0)$ "$<$ DOTTED"

STEP 1
FIND THE KEY POINTS

Your notes

FOR EACH GRAPH

$y = 2x + 1$

$(0,1)$ $(\frac{-1}{2},0)$ "$\geq$ SOLID"

$x + 2y = 3$

$(0,\frac{3}{2})$ $(3,0)$ "$>$ DOTTED"